

Function Operations			
	Add or Subtract Functions	Multiply or Divide Functions	Compose Functions
ALGEBRA	$(f + g)(x) = f(x) + g(x)$ $(f - g)(x) = f(x) - g(x)$	$(f \cdot g)(x) = f(x) \cdot g(x)$ $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$	$(f \circ g)(x) = f(g(x))$ $(g \circ f)(x) = g(f(x))$
WORDS	The domain of the sum or difference of f and g is the intersection of the domain of f and the domain of g .	The domain is the set of all real numbers for which f and g and the new function are defined.	The domain of $f \circ g$ is the set of all real numbers x , in the domain of g , such that $g(x)$ is in the domain of f .
NUMBERS	Let $f(x) = 3x + 5$ and $g(x) = x - 3$ $f + g = (3x + 5) + (x - 3) = 4x + 2$ $f - g = (3x + 5) - (x - 3) = 2x + 8$	Let $f(x) = 3x + 5$ and $g(x) = x - 3$ $f \cdot g = (3x + 5)(x - 3) = 3x^2 - 4x - 15$ $\frac{f}{g} = \frac{3x + 5}{x - 3}$ for $x \neq 3$	Let $f(x) = 3x + 5$ and $g(x) = x - 3$ $f \circ g = 3(x - 3) + 5 = 3x - 4$ $g \circ f = (3x + 5) - 3 = 3x + 2$